# Reading 7.3: Lab Results and Summary

We have introduced Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) as tools for image generation. VAEs aim to approximate the underlying data distribution P(X) using a latent distribution P(Z), which is assumed to be Gaussian. They utilize an Encoder-Decoder architecture, leveraging reconstruction loss to learn a latent space that enables new image generation. Additionally, the KL Divergence loss ensures P(Z) closely resembles a Gaussian distribution. In essence, VAEs are built upon two interconnected neural networks: the encoder, which maps input data to the latent space, and the decoder, which reconstructs the data from this latent representation.

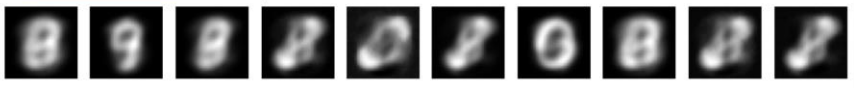
We also introduced Generative Adversarial Networks (GANs), an architecture designed to directly learn the underlying distribution of the training data, P(X). GANs comprise two neural networks: the generator, which creates images intended to deceive, and the discriminator, which aims to differentiate real images from the generated ones. These networks compete in a dynamic game, refining their outputs over time.

Next, we explored Conditional GANs, which add control over the generated outputs by conditioning both the generator and discriminator on auxiliary information, such as labels. Finally, we examined Conditional DCGANs, which replace dense layers with Convnets. Convnets excel at capturing spatial features, significantly enhancing image quality.

Here is a summary of the output performance of the 4 architectures sessions assuming 5 epochs and keeping all other parameters identic. Best results are for DC GAN and that is expected, followed by VAEs. The VAE results lack sharpness as expected. The digits in GAN and cGAN are not recognizable.

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Creating VAE model...
Training VAE...
Epoch 1/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 15s 27ms/step - kl_loss: 25.9592 - loss: 365.6115 - reconstruction_loss: 339.6523
Epoch 2/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 13s 28ms/step - kl_loss: 12.1766 - loss: 208.0875 - reconstruction_loss: 195.9108
Epoch 3/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 20s 27ms/step - kl_loss: 7.7454 - loss: 193.1144 - reconstruction_loss: 185.3690
Epoch 4/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 20s 27ms/step - kl_loss: 6.3098 - loss: 187.4461 - reconstruction_loss: 181.1362
Epoch 5/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 13s 28ms/step - kl_loss: 5.6783 - loss: 183.5098 - reconstruction_loss: 177.8315

Generating new digits...
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 64ms/step
```



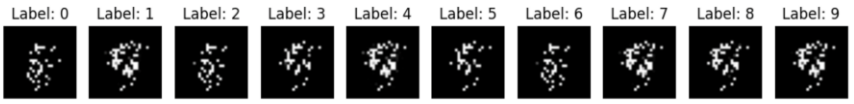
```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Creating GAN model...
Training GAN...
/usr/local/lib/python3.10/dist-packages/keras/src/layers/activations/leaky_relu.py:41: UserWarning: Argument `alpha`
  warnings.warn(
Epoch 1/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 43s 85ms/step - d_loss: 0.2196 - g_loss: 3.7425
Epoch 2/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 40s 83ms/step - d_loss: 0.1201 - g_loss: 4.2209
Epoch 3/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 40s 86ms/step - d_loss: 0.0744 - g_loss: 5.5796
Epoch 4/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 40s 83ms/step - d_loss: 0.0478 - g_loss: 5.5983
Epoch 5/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 40s 81ms/step - d_loss: 0.0183 - g_loss: 6.0197

Generating new digits...
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 67ms/step
```



```
Creating Conditional GAN model...
Training Conditional GAN...
Epoch 1/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 61s 123ms/step - d_loss: 0.2304 - g_loss: 3.6326
Epoch 2/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 54s 116ms/step - d_loss: 0.0672 - g_loss: 5.0429
Epoch 3/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 55s 117ms/step - d_loss: 0.0332 - g_loss: 5.5295
Epoch 4/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 55s 116ms/step - d_loss: 0.0110 - g_loss: 6.9872
Epoch 5/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 82s 117ms/step - d_loss: 0.0224 - g_loss: 7.9120

Generating digits...
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 87ms/step
```




```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 1s 0us/step
Creating Conditional DCGAN model...
/usr/local/lib/python3.10/dist-packages/keras/src/layers/activations/leaky_relu.py:41: UserWarning: Argument `alpha` is dep
  warnings.warn(
Training Conditional DCGAN...
Epoch 1/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 50s 64ms/step - d_loss: 0.4132 - g_loss: 2.1927
Epoch 2/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 18s 39ms/step - d_loss: 0.4957 - g_loss: 1.5004
Epoch 3/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 19s 40ms/step - d_loss: 0.6020 - g_loss: 1.1370
Epoch 4/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 19s 40ms/step - d_loss: 0.6266 - g_loss: 1.0066
Epoch 5/5
469/469 ━━━━━━━━━━━━━━━━━━━━ 19s 40ms/step - d_loss: 0.6187 - g_loss: 0.9843

Generating digits...
1/1 ━━━━━━━━━━━━━━━━━━━━ 1s 599ms/step
```

